

SELECTED WORK — CONTINUED

Native Max / MPE / Hardware Systems

Native C++ Max externals and hardware-integration work spanning multitouch, MPE, computer sensors, Arduino/ESP32 microcontrollers, MCP23017 I/O expansion, Depth Anything and AprilTag workflows. Includes Objective-C++ bridge layers, platform APIs, firmware integration/modification, native model/runtime integration and ongoing JUCE VST3/AU migration.

SoundCloner — Private R&D

Native synthesis and audio-analysis research platform combining deterministic DSP, oscillator/envelope/modulation systems, parameter/genome representations, similarity scoring, candidate retrieval, ML inference and large synthetic evaluation infrastructure.

Real-Time Target-Speaker Voice Isolation — Private R&D

Real-time target-speaker extraction using MossFormer2 + ECAPA, with speaker enrollment and suppression of competing voices.

Sample-Space Exploration — Private R&D

CLAP/audio embeddings, classifier training, similarity retrieval and Three.js spatial exploration of audio corpora, supported by benchmark and evaluation tooling.

iOS Computer Vision — Private R&D

Swift/SwiftUI iOS application work using ARKit, SceneKit, AVFoundation and MediaPipe Tasks Vision for real-time facial/eye landmark processing and guided physical-makeup workflows.

ENGINEERING VALIDATION

Repeatable benchmarking and regression methodology across projects: controlled baselines, latency/throughput profiling, held-out ML evaluation, parity testing, hardware-in-the-loop checks and cross-machine validation across Apple Silicon, Intel Mac and Windows execution paths where supported.

EXPERIENCE

Founder & Audio Software Developer — VSTOPIA | 2023–Present

Built an independent software ecosystem spanning product development, web/backend infrastructure, distribution, licensing, updates and developer workflows.

Independent Audio Software Developer — OSTIN SOLO | 2018–Present

Commercial and experimental audio technology across DSP, Max for Live, native Max externals, plug-ins, machine learning, computer vision, MPE/hardware control and networked music systems.

Founder, Artistic Director & Experience Designer — Macrowave / Cacao Discoteca / Basement | 2014–2016

Directed electronic-music and multimedia programmes in Italy for audiences up to 3,000, spanning artist curation, production, projection mapping, scenography and interactive installations.

EDUCATION

Music Production — SAE Institute | 2019–2020 · Recording, editing, mixing, mastering, sound design and synthesis

Electronic Music & Musicianship — Italian Conservatory | c. 2004–2009

RECOGNITION & PUBLIC EVIDENCE

Sound On Sound coverage of DSm · Cycling '74 official project listing · OSTIN SOLO commercial product catalogue · public AUDIOENCE platform and technical documentation.

Full technical portfolio: ostinsolo.github.io/ostin-solo-portfolio/